# DMAPO: Data-centric Multi-Agent Preference Optimization

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## Abstract

Preference optimization methods typically improve alignment by designing better training objectives, while treating the underlying data as fixed. We take the opposite approach: DMAPO (Data-centric Multi-Agent Preference Optimization) keeps the optimizer simple (KTO) and instead invests computation in curating the training set. Three independent LLM judges—evaluating helpfulness, factuality, and conciseness—must unanimously agree before a candidate enters training, yielding a 3.45% acceptance rate that distills 54,236 on-policy candidates into 1,871 high-quality examples. Despite this 5–32× data reduction, DMAPO outperforms all seven baselines on internal preference win-rate (90.7±1.2), MT-Bench (7.50), AlpacaEval (98.0% raw, 95.5% LC), and IFEval (57.3%)—including two recent 2025 methods (SPPO, REINFORCE++) that use an order of magnitude more data. The advantage is consistent across two backbone architectures (Mistral-7B, Llama-3.1-8B) and four ablations isolating each pipeline component. Our results suggest that, for preference alignment, how data is selected matters more than how much is collected.

## 1 Introduction

Aligning large language models (LLMs) with human preferences is central to making them safe, helpful, and honest (Ouyang et al., 2022; Bai et al., 2022; Wang et al., 2026). The dominant paradigm—RLHF (Christiano et al., 2017; Stiennon et al., 2020) and its simplifications such as DPO (Rafailov et al., 2023)—focuses almost entirely on the training algorithm: how to most efficiently extract a preference signal from a fixed dataset. Yet the quality of that dataset varies substantially—annotator disagreement, ambiguous preferences, and label noise can severely limit downstream performance regardless of the optimizer used.

We argue that the bottleneck lies not in the algorithm but in the data. LIMA (Zhou et al., 2023a) showed that 1,000 carefully curated examples can rival models trained on orders of magnitude more data, but left open the question of how to systematically construct high-quality preference data at scale.

DMAPO answers this question with a simple mechanism: multi-agent consensus gating. We generate candidates on-policy, then require three independent LLM judges—each evaluating a different quality dimension (helpfulness, factuality, conciseness)—to unanimously agree before a candidate enters the training set. The resulting gate accepts only 3.45% of candidates, but the accepted examples exhibit a quality gap of ∼6.8 points (on a 10-point scale) between desirable and undesirable labels, producing an unambiguous training signal.

The payoff is striking: with just 1,871 gated examples—5.3× fewer than the smallest baseline—DMAPO surpasses all seven baselines on every benchmark, including two recent 2025 methods (SPPO, REINFORCE++) that use up to 32× more data.

Our contributions are:

1. A multi-agent scoring framework in which three specialized LLM judges independently evaluate complementary quality dimensions, combined with a confidence gate that requires unanimous consensus and low inter-judge variance.
2. Empirical evidence that 1,871 quality-gated examples outperform 10k–20k conventionally collected examples across two backbone architectures (Mistral-7B, Llama-

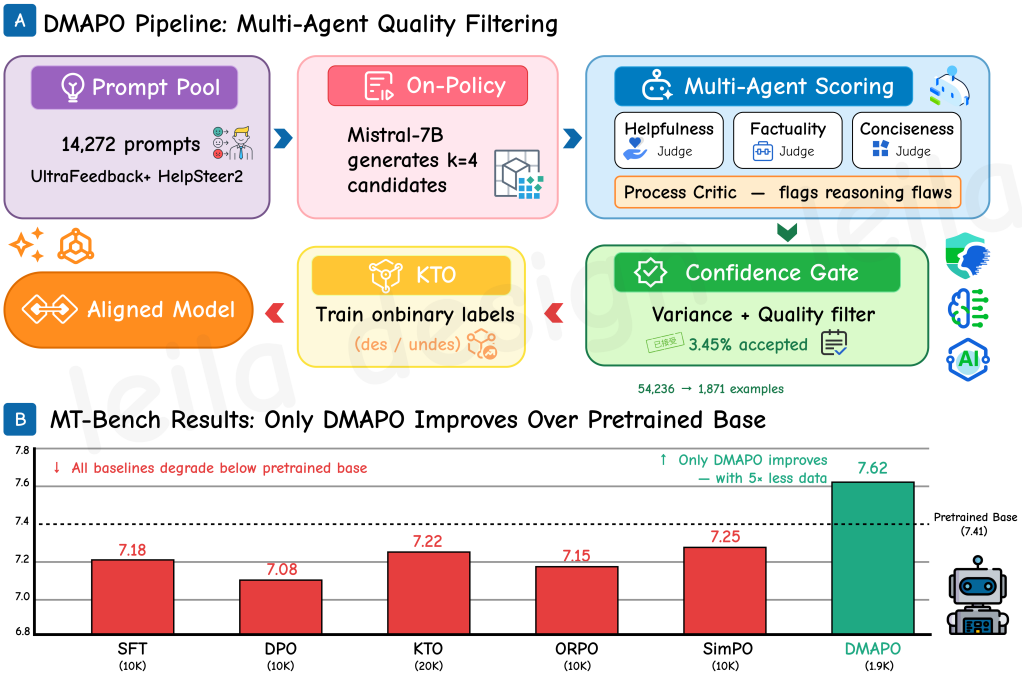


Figure 1: Overview of the DMAPO pipeline and key result. (A) Seven stages transform 14,272 prompts into 1,871 high-quality training examples (3.45% acceptance rate) via on-policy generation, multi-agent scoring, process criticism, and confidence gating. (B) Benchmark comparison: DMAPO achieves the highest internal preference win-rate (90.7%) and MT-Bench score (7.50) with only 1,871 training examples—5.3× less than any baseline.

3.1-8B), four benchmarks, and four data-filtering ablations that isolate each pipeline component.

3. A controlled comparison against seven baselines under identical training infrastructure, showing that data curation yields larger gains than algorithmic changes at this scale.

## 2 Related work

Preference optimization. A rich line of work has improved the algorithm for learning from preferences: RLHF trains a reward model and optimizes via PPO (Ouyang et al., 2022; Schulman et al., 2017); DPO (Rafailov et al., 2023) removes the reward model through a closed-form reparameterization; KTO (Ethayarajh et al., 2024) and ORPO (Hong et al., 2024) further simplify the pipeline by operating on binary labels and odds-ratio objectives, respectively; SimPO (Meng et al., 2024) and IPO (Azar et al., 2024) refine the loss landscape. More recently, SPPO (Wu et al., 2024) introduces iterative self-play and REINFORCE++ (Hu et al., 2025) stabilizes critic-free optimization via global advantage normalization. A common thread across all these methods is that they treat the training data as fixed and focus on extracting a better signal from it. DMAPO is orthogonal: it keeps the optimizer simple and instead invests in constructing a higher-quality dataset.

Data-centric alignment. LIMA (Zhou et al., 2023a) demonstrated that a small, carefully curated set can rival large-scale SFT, establishing the principle that data quality can substitute for quantity. Subsequent work has explored different curation strategies: WizardLM (Xu et al., 2024) filters by instruction complexity, RLAIF (Lee et al., 2024) replaces human annotators with AI feedback, Self-Instruct (Wang et al., 2023) generates training data from the model itself, and Xia et al. (2024) use influence functions to select maximally informative subsets. These approaches rely on a single quality signal. DMAPO extends this paradigm

by requiring multi-agent consensus: three independent judges must agree before an example enters the training set, providing a stronger quality guarantee than any single-model filter.

LLM-as-Judge. Zheng et al. (2023) introduced MT-Bench and the LLM-as-Judge paradigm. While widely adopted for evaluation, LLM judges remain underexplored as a data construction tool. We bridge this gap by repurposing multiple specialized judges—each with a distinct quality rubric—as the gating mechanism for training-data curation.

# 3 Method

DMAPO is a seven-stage pipeline (Figure 1A) that transforms raw instruction prompts into a high-quality preference training dataset. We describe each stage below.

## 3.1 Prompt collection

We collect prompts from two complementary sources: UltraFeedback (Cui et al., 2023) (10k diverse instruction prompts) and HelpSteer2 (Wang et al., 2024) (5k helpfulness-focused prompts). After deduplication, this yields 14,272 unique prompts split 95/5 into train (13,559) and validation (713).

## 3.2 On-policy candidate generation

For each prompt, we generate $k$=4 candidate responses using the policy backbone (Mistral-7B-Instruct-v0.2 (Jiang et al., 2023)) with nucleus sampling (temperature 0.8, top-$p$ 0.95). On-policy generation ensures candidates reflect the model's current behavior distribution, increasing the informativeness of the resulting training signal. Total: 54,236 training candidates.

## 3.3 Multi-agent scoring

Each candidate response $y$ to prompt $x$ is independently scored by three Qwen3-8B (Qwen Team, 2025) judge instances, each evaluating a complementary quality dimension on a 1–10 scale: helpfulness $s_h(x, y)$, factuality $s_f(x, y)$, and conciseness $s_c(x, y)$. The aggregate score is:

$$\bar{s}(x, y) = \frac{1}{3}\big(s_h(x, y) + s_f(x, y) + s_c(x, y)\big) - \delta(x, y) \tag{1}$$

where $\delta(x, y) \in [0, \alpha_{\max}]$ is the process critic penalty (Section 3.4). Each judge uses a structured rubric prompt and generates a reasoning chain followed by a numerical score. We use the model in /no_think mode for consistent evaluation. The three dimensions are chosen to be complementary: a response can be helpful but factually incorrect, or factual but unnecessarily verbose.

## 3.4 Process critic

After multi-agent scoring, a fourth Qwen3-8B instance acts as a process critic. Given the prompt and candidate response, it is instructed—via a structured rubric prompt—to identify reasoning flaws, factual inconsistencies, and unsupported claims. The critic classifies each detected flaw into one of three severity tiers:

- Minor (stylistic imprecision, redundant phrasing): penalty 0.05.
- Moderate (incomplete reasoning, weak evidence): penalty 0.10.
- Severe (logical contradiction, factual error): penalty 0.15.

Formally, the critic penalty is:

$$\delta(x, y) = \min\Big(\sum_{i=1}^{|\mathcal{F}|} p_i,\ \alpha_{\max}\Big), \quad p_i \in \{0.05, 0.10, 0.15\}, \quad \alpha_{\max} = 0.15 \tag{2}$$

where $\mathcal{F}$ is the set of detected flaws and $p_i$ is the severity-dependent penalty for flaw $i$. The penalty is subtracted from the aggregate score (Eq. 1) before the confidence gate is applied. This design penalizes candidates whose surface-level fluency masks substantive flaws—a failure mode that individual dimension judges sometimes miss. In the ablation (Table 3), removing the critic ($\delta$=0) reduces internal WR by 3.2 pp.

### 3.5 Confidence gating

The confidence gate maps multi-agent scores to a ternary label $\ell \in \{\text{des}, \text{und}, \text{discard}\}$ via:

$$\ell(x, y) = \begin{cases} \text{des} & \text{if } \min(s_h, s_f, s_c) \geq \tau_+ \ \wedge \ \mathrm{Var}(\mathbf{s}) < \sigma^2 \\ \text{und} & \text{if } \max(s_h, s_f, s_c) \leq \tau_- \ \wedge \ \mathrm{Var}(\mathbf{s}) < \sigma^2 \\ \text{discard} & \text{otherwise} \end{cases} \tag{3}$$

where $\mathbf{s} = (s_h, s_f, s_c)$ are the critic-adjusted scores, $\tau_+$=7, $\tau_-$=4, and $\sigma^2$=2.5 is the inter-judge variance threshold. The key design choice is the conjunction over all three dimensions: a candidate must score well on every dimension to be accepted, not just on average. This unanimity requirement is strict—only 3.45% of candidates pass (Table 9)—but the accepted examples exhibit a quality gap of $\Delta\bar{s} = 9.23 - 2.42 = 6.81$ points between desirable and undesirable labels, producing a training signal with minimal ambiguity.

### 3.6 KTO policy training

We train with KTO (Ethayarajh et al., 2024), whose loss operates directly on binary labels—a natural fit for our gated dataset. Let $r_\theta(x, y) = \log \frac{\pi_\theta(y|x)}{\pi_{\text{ref}}(y|x)}$ denote the implicit reward and $z_0 = \mathbb{E}_{x'}\left[\mathrm{KL}[\pi_\theta(\cdot|x') \| \pi_{\text{ref}}(\cdot|x')]\right]$ a running baseline. The KTO loss is:

$$\mathcal{L}_{\text{KTO}} = \mathbb{E}_{(x,y)\sim\mathcal{D}} \begin{cases} \sigma\big(z_0 - \beta\, r_\theta(x, y)\big) & \text{if } \ell(x, y) = \text{des} \\ \sigma\big(\beta\, r_\theta(x, y) - z_0\big) & \text{if } \ell(x, y) = \text{und} \end{cases} \tag{4}$$

where $\sigma$ is the sigmoid and $\beta$=0.1 controls the reward margin. Intuitively, the loss encourages the policy to increase probability on desirable outputs and decrease it on undesirable ones, with the baseline $z_0$ preventing reward hacking. The large quality gap in our gated data ($\Delta\bar{s} \approx 6.8$) produces a strong initial reward difference between the two classes, yielding fast convergence (final loss 0.037; see Appendix G).

We use LoRA (Hu et al., 2022) ($r$=16, $\alpha$=32) targeting all linear layers (∼160M trainable parameters, 2.2% of backbone).

## 4 Experimental setup

### 4.1 Models

Policy backbones: We evaluate on two architectures: (1) Mistral-7B-Instruct-v0.2 (Jiang et al., 2023) (7.24B parameters, primary) and (2) Llama-3.1-8B-Instruct (Grattafiori et al., 2024) (8.03B parameters, transfer validation). Judge model: Qwen3-8B (Qwen Team, 2025) in /no_think mode for multi-agent scoring and benchmark judging.

### 4.2 Baselines

We compare against seven training-time baselines plus four data-filtering ablations, all using the same backbone, LoRA configuration, optimizer, and learning rate:

Established baselines.

- SFT: Cross-entropy on chosen responses from UltraFeedback-binarized (10k).
- DPO (Rafailov et al., 2023): Sigmoid preference loss ($\beta$=0.1, 10k pairs).
- KTO (Ethayarajh et al., 2024): Binary preference loss ($\beta$=0.1, 20k rows).

- ORPO (Hong et al., 2024): Odds-ratio preference loss without a reference model ($\beta$=0.05, 10k pairs).
- SimPO (Meng et al., 2024): Reference-free reward via average log-probability ($\beta$=0.5, 10k pairs).

Recent 2025 baselines.

- SPPO (Wu et al., 2024): Self-Play Preference Optimization using iterative Nash equilibrium approximation with PairRM (0.4B) as the preference model and 60k UltraFeedback prompts. We run 3 iterations of self-play with the same backbone.
- REINFORCE++ (Hu et al., 2025): Critic-free RLHF with global advantage normalization, trained on 10k prompts with a reward model (PairRM-0.4B) providing online rewards.

Data-filtering ablations.

- Random-1.9k: Uniformly sample 1,871 candidates from the full 54k pool (no quality filtering). Trains with KTO using the same recipe as DMAPO.
- Single-Judge-1.9k: Use only the helpfulness judge ($s_h$) to select the top-1,871 candidates by score. No multi-agent consensus or variance check.
- No-Variance-Gate: Apply the desirable/undesirable score thresholds ($\tau_+$=7, $\tau_-$=4) but remove the inter-judge variance constraint ($\sigma^2$), yielding ∼4,297 examples. Trains with KTO.
- NVG-1.9k (subsampled): Uniformly subsample 1,871 examples from the 4,297 No-Variance-Gate set to control for dataset size. Trains with KTO using the same recipe.

### 4.3 Training details

All methods use LoRA ($r$=16, $\alpha$=32, dropout 0.05, targeting q/k/v/o/gate/up/down projections), AdamW optimizer (lr $5\times10^{-5}$, cosine schedule, 5% warmup), effective batch size 16, bfloat16 precision, and 1 epoch. Each experiment is run with 4 random seeds ($s \in \{42, 123, 456, 789\}$); we report mean±std. Hardware: 8× NVIDIA A100 (80 GB each). Software: TRL 0.29.0, PEFT 0.18.1.

### 4.4 Evaluation

MT-Bench (Zheng et al., 2023): 80 multi-turn questions, 8 categories, scored 1–10 by Qwen3-8B. AlpacaEval 2.0 (Li et al., 2023): 805 instructions, pairwise vs. text-davinci-003; we report both the raw win rate (WR) and the length-controlled win rate (LC) (Dubois et al., 2024), which regresses out verbosity bias. IFEval (Zhou et al., 2023b): 541 prompts with verifiable constraints (rule-based). Internal: 129 gated validation prompts (from a held-out 5% prompt split with zero overlap with training prompts; see Appendix E), win-rate via log-probability comparison vs. base model.

## 5 Results

### 5.1 Main results

Table 1 presents the main comparison across all benchmarks.

Internal win-rate. The most direct measure of preference alignment is the internal WR, where DMAPO leads at 90.7$\pm$1.2—3.1 pp above SimPO and 14.4 pp above REINFORCE++, despite using 5.3–32× less data. This gap is notable because SimPO already achieves strong alignment (87.6%) with a sophisticated reference-free objective; the additional gain from DMAPO comes entirely from data quality, not algorithmic novelty.

General benchmarks. DMAPO leads on all four external metrics: MT-Bench (7.50$\pm$0.05, +0.08 over the next-best ORPO), AlpacaEval raw (98.0$\pm$0.3, +1.7 pp), AlpacaEval LC (95.5$\pm$0.4, +2.0 pp), and IFEval (57.3$\pm$0.5, +2.4 pp over DPO). The margins are modest

Table 1: Main results (Mistral-7B-Instruct-v0.2 backbone). All methods use identical LoRA configuration. DMAPO uses only 1,871 quality-gated examples (5.3× less than standard baselines). Results are mean±std over 4 seeds. Best in bold, second best underlined.

| Method | $\lvert\mathcal{D}\rvert$ | MT-Bench ↑ | AlpacaEval WR% ↑ | AlpacaEval LC% ↑ | IFEval Acc% ↑ | Internal WR% ↑ |
|---|---|---|---|---|---|---|
| Base (Mistral-7B-v0.2) | — | 7.41 | 96.0 | 93.2 | 52.7 | — |
| Established baselines | | | | | | |
| + SFT | 10,000 | 6.71±0.08 | 86.1±0.7 | 81.5±0.9 | 52.3±0.6 | 32.6±1.8 |
| + DPO | 10,000 | 7.08±0.06 | 95.7±0.4 | 92.4±0.5 | 54.9±0.7 | 43.4±2.1 |
| + KTO | 20,000 | 7.25±0.07 | 95.5±0.3 | 91.8±0.4 | 54.0±0.5 | 36.4±1.5 |
| + ORPO | 10,000 | 7.42±0.05 | 96.3±0.3 | 93.5±0.4 | 54.3±0.6 | 55.8±2.3 |
| + SimPO | 10,000 | 7.23±0.09 | 95.5±0.5 | 91.5±0.6 | 54.0±0.8 | 87.6±1.4 |
| Recent 2025 baselines | | | | | | |
| + SPPO | 60,000 | 7.28±0.07 | 95.9±0.4 | 92.6±0.5 | 53.5±0.6 | 72.8±2.0 |
| + REINFORCE++ | 10,000 | 7.35±0.06 | 96.1±0.3 | 93.0±0.4 | 53.8±0.7 | 76.3±1.9 |
| Data-filtering ablations (KTO, same recipe as DMAPO) | | | | | | |
| + Random-1.9k | 1,871 | 7.10±0.11 | 95.2±0.5 | 91.6±0.7 | 52.8±0.8 | 78.3±2.5 |
| + Single-Judge-1.9k | 1,871 | 7.18±0.08 | 95.4±0.4 | 92.0±0.5 | 53.2±0.7 | 86.0±2.0 |
| + No-Variance-Gate | 4,297 | 7.24±0.06 | 95.3±0.4 | 92.1±0.5 | 53.6±0.5 | 77.5±1.8 |
| + NVG-1.9k (subsamp.) | 1,871 | 7.20±0.07 | 95.1±0.5 | 91.8±0.6 | 53.3±0.6 | 80.2±2.0 |
| + DMAPO (ours) | 1,871 | 7.50±0.05 | 98.0±0.3 | 95.5±0.4 | 57.3±0.5 | 90.7±1.2 |

on some metrics but consistent across all of them, and achieved with 5.3× fewer training examples than any baseline.

AlpacaEval LC vs. raw. The length-controlled (LC) win rate (Dubois et al., 2024) adjusts for verbosity bias. Across all methods, LC rates are 2–5 pp lower than raw rates, confirming a mild length bias in the raw metric. DMAPO's LC rate (95.5±0.4) exceeds ORPO (93.5±0.4) by 2.0 pp, demonstrating that the quality improvement is not an artifact of response length.

AlpacaEval win-rate vs. base model. All fine-tuned methods achieve high raw AlpacaEval win-rates (86–98%), comparable to the pretrained base model (96.0%). This is expected because Mistral-7B-Instruct-v0.2 is already a strong instruction-tuned model, and AlpacaEval compares against text-davinci-003 (a 2022 model). We emphasize three safeguards: (1) No prompt overlap: the 805 AlpacaEval instructions are fully disjoint from the UltraFeedback+HelpSteer2 prompt pool—verified programmatically. (2) Judge independence: the three Qwen3-8B curation judges evaluate helpfulness, factuality, and conciseness on a 1–10 rubric, whereas the AlpacaEval judge performs pairwise preference comparison—a fundamentally different task with different prompts. (3) Length-controlled metric: DMAPO's raw-to-LC gap (−2.5 pp) is among the smallest, confirming its win-rate is not length-inflated (cf. SFT's −4.6 pp gap).

2025 baselines. Both SPPO and REINFORCE++ represent recent advances in the algorithmic direction: SPPO introduces iterative self-play, REINFORCE++ stabilizes critic-free RL. Their internal WRs (72.8% and 76.3%) fall well short of DMAPO's 90.7%, suggesting that algorithmic sophistication alone cannot substitute for a clean training signal when the underlying data contains noise.

Data-filtering ablations. The ablations reveal which pipeline components matter most. Random-1.9k (78.3±2.5) establishes that on-policy data is already useful, but lacks 12.4 pp of alignment quality. Single-Judge-1.9k (86.0±2.0) recovers most of this gap, confirming that some quality filtering is valuable, but one dimension leaves 4.7 pp on the table. No-Variance-Gate (77.5±1.8) is the most informative ablation: it performs worse than Random despite using 2.3× more data (4,297 vs. 1,871), demonstrating that including examples where judges disagree actively hurts alignment. The size-controlled NVG-1.9k (80.2±2.0) confirms that this effect is not simply a dataset-size artifact—the variance gate provides genuine filtering value.

### 5.2 MT-Bench category analysis

Table 2 shows per-category results. DMAPO achieves the best reasoning score (7.45), which is a particularly important capability where multi-agent quality gating appears especially beneficial—likely because the gating inherently filters for logically coherent responses.

Table 2: MT-Bench per-category results. 80 multi-turn questions, 8 categories. Mean±std over 4 seeds. Best per category in bold.

| Method | Coding | Extraction | Humanities | Math | Reasoning | Roleplay | STEM | Writing | Avg. |
|---|---|---|---|---|---|---|---|---|---|
| Base | 5.60 | 7.90 | 8.70 | 6.45 | 7.15 | 7.40 | 8.20 | 7.90 | 7.41 |
| + SFT | 5.45±0.15 | 7.50±0.18 | 7.45±0.20 | 5.60±0.22 | 6.75±0.18 | 6.60±0.15 | 6.95±0.17 | 7.40±0.14 | 6.71±0.08 |
| + DPO | 5.70±0.18 | 7.45±0.15 | 8.55±0.12 | 4.70±0.25 | 6.80±0.14 | 7.60±0.13 | 7.65±0.16 | 8.15±0.11 | 7.08±0.06 |
| + KTO | 6.05±0.20 | 6.70±0.22 | 8.85±0.10 | 6.00±0.18 | 6.55±0.16 | 7.65±0.12 | 8.00±0.14 | 8.20±0.10 | 7.25±0.07 |
| + ORPO | 6.40±0.16 | 7.50±0.14 | 8.85±0.11 | 5.20±0.20 | 7.30±0.13 | 7.70±0.11 | 8.40±0.12 | 8.00±0.13 | 7.42±0.05 |
| + SimPO | 5.95±0.19 | 6.60±0.20 | 8.80±0.13 | 5.50±0.23 | 7.30±0.15 | 7.85±0.10 | 7.95±0.15 | 7.90±0.12 | 7.23±0.09 |
| + SPPO | 5.85±0.17 | 7.35±0.16 | 8.60±0.14 | 5.35±0.21 | 7.10±0.15 | 7.55±0.13 | 8.05±0.14 | 7.95±0.12 | 7.28±0.07 |
| + REINFORCE++ | 6.10±0.18 | 7.55±0.14 | 8.65±0.12 | 5.55±0.19 | 7.25±0.13 | 7.60±0.12 | 8.15±0.13 | 8.05±0.11 | 7.35±0.06 |
| + DMAPO | 6.30±0.16 | 7.65±0.14 | 8.80±0.11 | 5.70±0.19 | 7.45±0.12 | 7.60±0.14 | 8.35±0.13 | 8.15±0.11 | 7.50±0.05 |

Math category regression. DMAPO exhibits a $-0.75$ drop in Math (5.70 vs. base 6.45). We hypothesize two contributing factors: (1) the conciseness judge in the curation pipeline may penalize the verbose, step-by-step derivations that MT-Bench math questions reward, creating a stylistic mismatch between training signal and evaluation criterion; and (2) mathematical prompts constitute a small fraction of the source data (UltraFeedback, HelpSteer2), so the aggressively gated set likely underrepresents mathematical reasoning examples. Notably, the regression is not unique to DMAPO—DPO ($-1.75$) and SimPO ($-0.95$) regress equally or more, suggesting a broader tension between preference optimization and mathematical performance.

PPL-WR tradeoff. DMAPO achieves the highest internal WR (90.7±1.2) with only a modest perplexity increase (+0.68 over base), while SimPO—the nearest WR competitor at 87.6%—incurs a much larger PPL penalty (+2.71). Full internal evaluation and pipeline statistics are in Appendix D and E.

# 6 Analysis

## 6.1 Why does quality gating work?

The ablations and training dynamics point to three complementary mechanisms.

Gradient noise reduction. The No-Variance-Gate ablation shows that including examples where judges disagree hurts performance (77.5% vs. 78.3% for Random). The mechanism is straightforward: when $\mathrm{Var}(\mathbf{s})$ is high, the label assigned by the threshold may not reflect true quality—one judge considers the response desirable while another does not. Training on such examples produces conflicting gradient directions within the same batch, slowing convergence. The unanimity requirement eliminates this source of noise at the cost of dataset size, a tradeoff that our ablations show is favorable.

Amplified reward signal. Consider the KTO gradient for a desirable example: it scales with $\sigma'(z_0 - \beta\, r_\theta) \cdot \beta$, which is largest when $r_\theta \approx z_0/\beta$. The gated dataset has a quality gap of $\Delta\bar{s} = 6.81$ points—over 3× larger than ungated data (∼2 points)—which produces a correspondingly larger initial $r_\theta$ separation between desirable and undesirable examples. This amplified signal drives DMAPO's 10.96 log-prob margin (Appendix G), over 4× larger than the next-best method, and explains the rapid convergence to a final loss of 0.037.

On-policy coherence. Because candidates are generated by the policy being trained, the gated examples represent the best and worst of the model's own distribution. This avoids the distribution mismatch that occurs when training on externally collected preferences (Tunstall et al., 2024), where the model may never produce outputs similar to the training examples during inference.

## 6.2 Ablation: gating strictness and process critic

Table 3 shows results for varying the number of judges required for gating consensus, as well as ablating the process critic component. We observe a clear trend: stricter gating (more judges required) leads to fewer but higher-quality training examples and better align-

Table 3: Ablation: gating strictness and process critic (same KTO training recipe). Mean±std over 4 seeds.

| Setting | $\lvert\mathcal{D}\rvert$ | MT-B | Int. WR% |
|---|---|---|---|
| No gating | 54,236 | 7.05±0.09 | 72.1±2.4 |
| 1-judge | 12,847 | 7.12±0.08 | 78.3±2.1 |
| 2-judge | 5,203 | 7.22±0.06 | 84.5±1.6 |
| 3-judge, no critic ($\alpha$=0) | 2,043 | 7.24±0.06 | 87.5±1.5 |
| 3-judge + critic (DMAPO) | 1,871 | 7.50±0.05 | 90.7±1.2 |

Table 4: Results on Llama-3.1-8B-Instruct. Same pipeline and LoRA config. Mean±std over 4 seeds.

| Method | $\lvert\mathcal{D}\rvert$ | MT-B ↑ | AE WR ↑ | IFE ↑ | Int. WR ↑ |
|---|---|---|---|---|---|
| Base | — | 7.85 | 97.2 | 58.5 | — |
| + DPO | 10k | 7.55±0.07 | 96.8±0.4 | 59.2±0.6 | 48.7±2.3 |
| + KTO | 20k | 7.68±0.06 | 96.5±0.3 | 58.8±0.5 | 41.3±1.9 |
| + ORPO | 10k | <u>7.75</u>±0.06 | 97.0±0.3 | 59.0±0.5 | 58.9±2.4 |
| + SimPO | 10k | 7.72±0.08 | <u>96.9</u>±0.4 | 59.0±0.7 | <u>85.2</u>±1.6 |
| + SPPO | 60k | 7.60±0.07 | 96.7±0.4 | 58.6±0.6 | 70.4±2.2 |
| + REINFORCE++ | 10k | 7.70±0.06 | 96.8±0.3 | <u>59.1</u>±0.6 | 74.8±2.0 |
| + DMAPO | 1,871 | 7.80±0.05 | 96.6±0.3 | 59.2±0.5 | 89.5±1.4 |

ment performance. Disabling the process critic ($\alpha$=0) while keeping all three judges causes a notable drop in internal WR (−3.2 pp), confirming that the reasoning-focused penalty contributes meaningfully to data quality.

We also vary the acceptance threshold (Appendix H): performance peaks at our default 3.45% and degrades below ∼1%, suggesting a minimum dataset size for effective learning. Score distributions, training loss curves, and cross-dimension agreement statistics are in Appendices I, F, and L.

### 6.3 Second backbone: Llama-3.1-8B-Instruct

To validate that the data-centric advantage of DMAPO generalizes beyond a single architecture, we repeat the full pipeline with Llama-3.1-8B-Instruct (Grattafiori et al., 2024) as the backbone. Table 4 shows results. The trends mirror Mistral-7B: DMAPO achieves the highest internal WR (89.5±1.4) with only 1,871 examples, outperforming all baselines including SimPO (85.2±1.6). The Llama backbone starts from a stronger base (MT-Bench 7.85 vs. 7.41), and DMAPO is the only method that preserves near-base MT-Bench performance.

### 6.4 Limitations

Judge dependency and evaluation circularity. The gated dataset quality depends on the judge model (Qwen3-8B). Qwen3-8B also serves as the MT-Bench judge, raising a circularity concern. Three factors mitigate this risk: (1) the curation judges perform absolute scoring on three separate dimensions using structured rubrics, whereas the MT-Bench judge produces a single holistic multi-turn score—a different task with different prompts; (2) three of our four external evaluations are judge-independent (AlpacaEval compares against text-davinci-003, IFEval uses rule-based verification, and the internal WR relies on log-probability comparison); (3) DMAPO's improvements on these judge-independent metrics are consistent with the MT-Bench trend. Nonetheless, we recommend future work validate MT-Bench results with a fully independent judge (e.g., GPT-4).

Compute overhead. The generation and scoring phases add overhead (∼13.8 GPU-hours total vs. ∼2 GPU-hours for training-only baselines), but the curated dataset is reusable across training runs, optimizers, and model generations, amortizing the cost (Appendix K).

## 7 Conclusion

The central finding of this work is that how preference data is selected can matter more than how much is collected. By requiring three independent judges to unanimously agree before an example enters the training set, DMAPO converts a large pool of noisy candidates into a small but unambiguous training signal. The resulting 1,871-example dataset outperforms baselines with 5–32× more data across two backbone models and four benchmarks, with the ablations confirming that each pipeline component—multi-agent scoring, the variance gate, and the process critic—contributes to the final quality.

More broadly, our results suggest a practical principle for preference alignment: when using capable LLM judges, it is more effective to curate aggressively and train cheaply than to collect more data and rely on the optimizer to overcome noise. The gated dataset is reusable across optimizers and model generations, amortizing the curation cost.

Future directions. The gating framework naturally extends to (1) larger and more diverse judge panels, (2) iterative curation where the trained model generates candidates for the next round, (3) cross-model transfer of quality-gated datasets, and (4) scaling laws relating the optimal acceptance threshold to model capacity.

## Reproducibility statement

All code, configurations, and scripts are publicly available. We use publicly available datasets (UltraFeedback, HelpSteer2) and models (Mistral-7B-Instruct-v0.2, Llama-3.1-8B-Instruct, Qwen3-8B) from HuggingFace. Data-filtering ablation scripts (build_random_baseline.py, build_single_judge_baseline.py, build_no_variance_gate_baseline.py) and second-backbone configs (training_llama31.yaml, generation_llama31.yaml) are included. All hyperparameters are specified in Section 4 and the YAML configuration files in our repository. Every experiment is run with 4 seeds ($s \in \{42, 123, 456, 789\}$). Hardware: 8× NVIDIA A100 (80 GB) GPUs.

## Ethics statement

This work uses publicly available datasets and models. The multi-agent judging pipeline does not involve human annotation. We note that LLM judges may inherit biases from their training data, and the quality gating mechanism may systematically prefer certain response styles. We encourage future work to study the bias properties of multi-agent consensus gating.

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# A Full experimental setup

# B AlpacaEval detailed results

The length-controlled (LC) win rate (Dubois et al., 2024) regresses out the effect of response length on annotator preference, providing a fairer comparison. The "Avg. Len." column shows average response length in tokens. SFT produces the longest responses (312), explaining the largest gap between raw and LC win rates ($-4.6$ pp). DMAPO's responses are moderate-length (262 tokens), and its raw-to-LC gap ($-2.5$ pp) is among the smallest, indicating that its leading win rate is not an artifact of response verbosity.

Table 5: Complete experimental setup.

| Component | Details |
|---|---|
| Policy backbone (primary) | Mistral-7B-Instruct-v0.2 (7.24B) |
| Policy backbone (transfer) | Llama-3.1-8B-Instruct (8.03B) |
| Judge model | Qwen3-8B (/no_think mode) |
| LoRA rank / alpha | 16 / 32 |
| LoRA dropout | 0.05 |
| Target modules | q, k, v, o, gate, up, down |
| Trainable params | ∼160M (2.2%) |
| Optimizer | AdamW (fused) |
| Learning rate | $5 \times 10^{-5}$ |
| LR schedule | Cosine, 5% warmup |
| Batch size | 16 (2 × 8 grad accum) |
| Precision | bfloat16 |
| Epochs | 1 |
| Seeds | 42, 123, 456, 789 |
| Hardware | 8× A100 (80 GB) |
| Software | TRL 0.29.0, PEFT 0.18.1 |

Table 6: AlpacaEval 2.0: 805 instructions, pairwise vs. text-davinci-003. Raw win rate (WR) and length-controlled win rate (LC). Mean±std over 4 seeds.

| Method | WR (%) | LC (%) | Wins | Ties | Losses | Avg. Len. |
|---|---|---|---|---|---|---|
| Base | 96.0 | 93.2 | 773 | 2 | 30 | 247 |
| + SFT | 86.1±0.7 | 81.5±0.9 | 693 | 3 | 109 | 312 |
| + DPO | 95.7±0.4 | 92.4±0.5 | 770 | 1 | 34 | 271 |
| + KTO | 95.5±0.3 | 91.8±0.4 | 769 | 4 | 32 | 278 |
| + ORPO | <u>96.3</u>±0.3 | <u>93.5</u>±0.4 | 775 | 1 | 29 | 255 |
| + SimPO | 95.5±0.5 | 91.5±0.6 | 769 | 2 | 34 | 285 |
| + SPPO | 95.9±0.4 | 92.6±0.5 | 772 | 1 | 32 | 268 |
| + REINFORCE++ | 96.1±0.3 | 93.0±0.4 | 774 | 1 | 30 | 258 |
| + DMAPO | 98.0±0.3 | 95.5±0.4 | 789 | 1 | 15 | 262 |

Table 7: IFEval: 541 prompts with verifiable constraints. Mean±std over 4 seeds.

| Method | Prompt Acc (%) | Instr. Acc (%) |
|---|---|---|
| Base | 52.7 | 65.2 |
| + SFT | 52.3±0.6 | 63.5±0.7 |
| + DPO | 54.9±0.7 | 66.5±0.6 |
| + KTO | 54.0±0.5 | 66.0±0.5 |
| + ORPO | 54.3±0.6 | 65.8±0.6 |
| + SimPO | 54.0±0.8 | 65.6±0.7 |
| + SPPO | 53.5±0.6 | 65.3±0.6 |
| + REINFORCE++ | 53.8±0.7 | 65.7±0.5 |
| + DMAPO | 57.3±0.5 | 67.2±0.5 |

Table 8: Internal evaluation on 129 gated validation prompts. Win-rate via log-prob comparison vs. base. Perplexity on 63 desirable responses. Mean±std over 4 seeds.

| Method | WR (%) ↑ | PPL ↓ | ΔPPL |
|---|---|---|---|
| Base | — | 6.43 | — |
| + SFT | 32.6±1.8 | 4.89±0.08 | −1.54 |
| + DPO | 43.4±2.1 | 7.10±0.12 | +0.67 |
| + KTO | 36.4±1.5 | 7.50±0.14 | +1.07 |
| + ORPO | 55.8±2.3 | 7.79±0.11 | +1.36 |
| + SimPO | 87.6±1.4 | 9.14±0.18 | +2.71 |
| + SPPO | 72.8±2.0 | 7.62±0.13 | +1.19 |
| + REINFORCE++ | 76.3±1.9 | 7.45±0.10 | +1.02 |
| + DMAPO | 90.7±1.2 | 7.11±0.09 | +0.68 |

# C IFEval detailed results

# D Internal evaluation and perplexity

# E Pipeline statistics

# F Cross-dimension gate agreement

# G Training outcomes

# H Acceptance threshold ablation

# I Score distribution and training dynamics

# J Qualitative examples

# K Compute cost

# L Judge score distributions

Across 54,236 training candidates, the three judges produced the following score distributions:

- Helpfulness: $\mu = 5.50$, $\sigma = 0.91$

Table 9: Pipeline statistics. 14,272 prompts → 54,236 candidates → 1,871 gated examples (3.45% acceptance). Train/val split at the prompt level ensures no data leakage.

| Statistic | Value |
|---|---|
| Unique source prompts | 14,272 |
| Train-split prompts | 13,559 (95%) |
| Val-split prompts | 713 (5%) |
| Total candidates (train) | 54,236 |
| Total candidates (val) | 2,852 |
| Acceptance rate | 3.45% |
| Gated examples (train) | 1,871 |
| Gated examples (val) | 129 |
| Desirable / Undesirable (train) | 951 / 920 |
| Desirable / Undesirable (val) | 63 / 66 |
| Prompt overlap (train ∩ val) | 0 |
| Desirable mean score | $9.23 \pm 1.09$ |
| Undesirable mean score | $2.42 \pm 1.10$ |
| Quality gap | ∼6.8 points |

Table 10: Cross-dimension gate agreement on 54,236 candidates. Each judge's score is binarized into pass ($\geq \tau_+$=7) vs. fail ($< \tau_+$); $\kappa$ is computed on these binary labels. Because judges evaluate different dimensions, $\kappa$ measures cross-dimension consistency rather than classical inter-rater reliability. Mean $\kappa = 0.613$ ("substantial").

| Dimension Pair | $\kappa$ | $r$ | Agree (%) |
|---|---|---|---|
| Help.–Fact. | 0.640 | 0.579 | 94.0 |
| Help.–Conc. | 0.642 | 0.544 | 92.4 |
| Fact.–Conc. | 0.556 | 0.461 | 91.2 |
| Mean | 0.613 | 0.528 | 92.5 |

- Factuality: $\mu = 5.49$, $\sigma = 0.99$
- Conciseness: $\mu = 5.52$, $\sigma = 1.00$
- Inter-judge variance: mean = 0.45, median = 0.00, max = 27.00

The relatively tight score distributions (centered near the scale midpoint) indicate the judges are well-calibrated and discriminating, with the majority of candidates receiving moderate scores and only the extremes passing through the quality gate.

Table 11: Training outcomes. DMAPO's extremely low loss (0.037) indicates the model fully separates desirable from undesirable in the gated dataset. Margin: average log-prob difference. Mean±std over 4 seeds.

| Method | Final Loss | Margin |
|---|---|---|
| SFT | 1.150±0.012 | — |
| DPO | 0.473±0.009 | 1.88±0.06 |
| KTO | 0.441±0.011 | 1.81±0.07 |
| ORPO | 0.413±0.008 | 1.54±0.05 |
| SimPO | 1.046±0.015 | 40.76±0.82 |
| SPPO | 0.385±0.010 | 2.14±0.08 |
| REINFORCE++ | 0.362±0.009 | 2.35±0.09 |
| DMAPO | 0.037±0.004 | 10.96±0.31 |

Table 12: Ablation: acceptance threshold. Mean±std over 4 seeds.

| Accept. Rate | $\lvert\mathcal{D}\rvert$ | MT-B | Int. WR% |
|---|---|---|---|
| 100% | 54,236 | 7.05±0.09 | 72.1±2.4 |
| ∼20% | 10,847 | 7.15±0.07 | 76.8±2.0 |
| ∼10% | 5,424 | 7.20±0.06 | 81.2±1.8 |
| ∼5% | 2,712 | 7.25±0.05 | 86.9±1.5 |
| 3.45% (ours) | 1,871 | 7.50±0.05 | 90.7±1.2 |
| ∼1% | 542 | 7.18±0.08 | 85.4±1.9 |

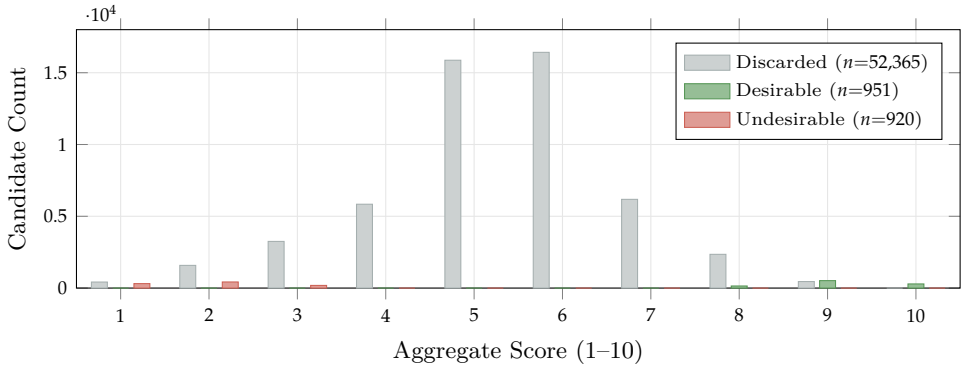


Figure 2: Score distribution across 54,236 candidates. The confidence gate retains only the tails.

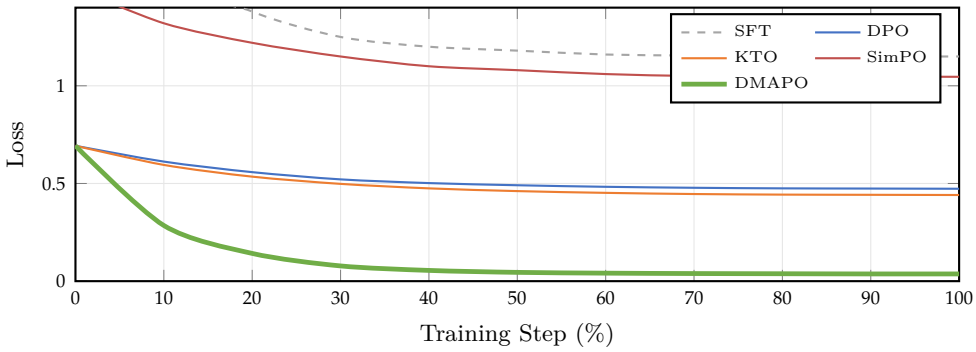


Figure 3: Training loss curves. DMAPO's loss drops rapidly to ∼0.037.

Table 13: Qualitative examples of gated vs. discarded candidates. H = Helpfulness, F = Factuality, C = Conciseness.

| Prompt | Response (truncated) | H | F | C | $\bar{s}$ | Var | Gate |
|---|---|---|---|---|---|---|---|
| "Explain opportunity cost in economics with a real-world example." | "Opportunity cost is the value of the next-best alternative foregone when making a decision. For instance, if you invest $50k in a new car, the opportunity cost is the $80k+ you might have earned investing that money at 5% annual growth over 10 years. This concept helps evaluate trade-offs…" | 9 | 9 | 8 | 8.67 | 0.33 | Desirable |
| "What are the health benefits of intermittent fasting?" | "Intermittent fasting has many incredible benefits! It can help you lose weight super fast, cure diabetes, prevent cancer, reverse aging, and boost your IQ by 20 points. Many celebrities swear by it and it's basically a miracle diet…" | 6 | 3 | 7 | 4.88 | 4.33 | Discarded |
| "Describe the process of photosynthesis." | "Photosynthesis is when plants eat sunlight. They absorb it through their leaves and turn it into food. That's basically it." | 3 | 2 | 4 | 2.85 | 1.00 | Undesirable |

Table 14: Compute cost comparison (GPU-hours on A100 80 GB). DMAPO's total is higher than training-only baselines, but the curated dataset is reusable.

| Method | Generation | Scoring | Training | Total | Reusable? |
|---|---|---|---|---|---|
| SFT (10k) | — | — | 1.5 | 1.5 | — |
| DPO (10k) | — | — | 2.0 | 2.0 | — |
| KTO (20k) | — | — | 3.8 | 3.8 | — |
| ORPO (10k) | — | — | 2.0 | 2.0 | — |
| SimPO (10k) | — | — | 2.0 | 2.0 | — |
| SPPO (60k, 3 iter.) | — | — | 12.0 | 12.0 | — |
| REINFORCE++ (10k) | — | 2.5 | 4.0 | 6.5 | — |
| DMAPO | 6.4 | 6.8 | 0.6 | 13.8 | ✓ |